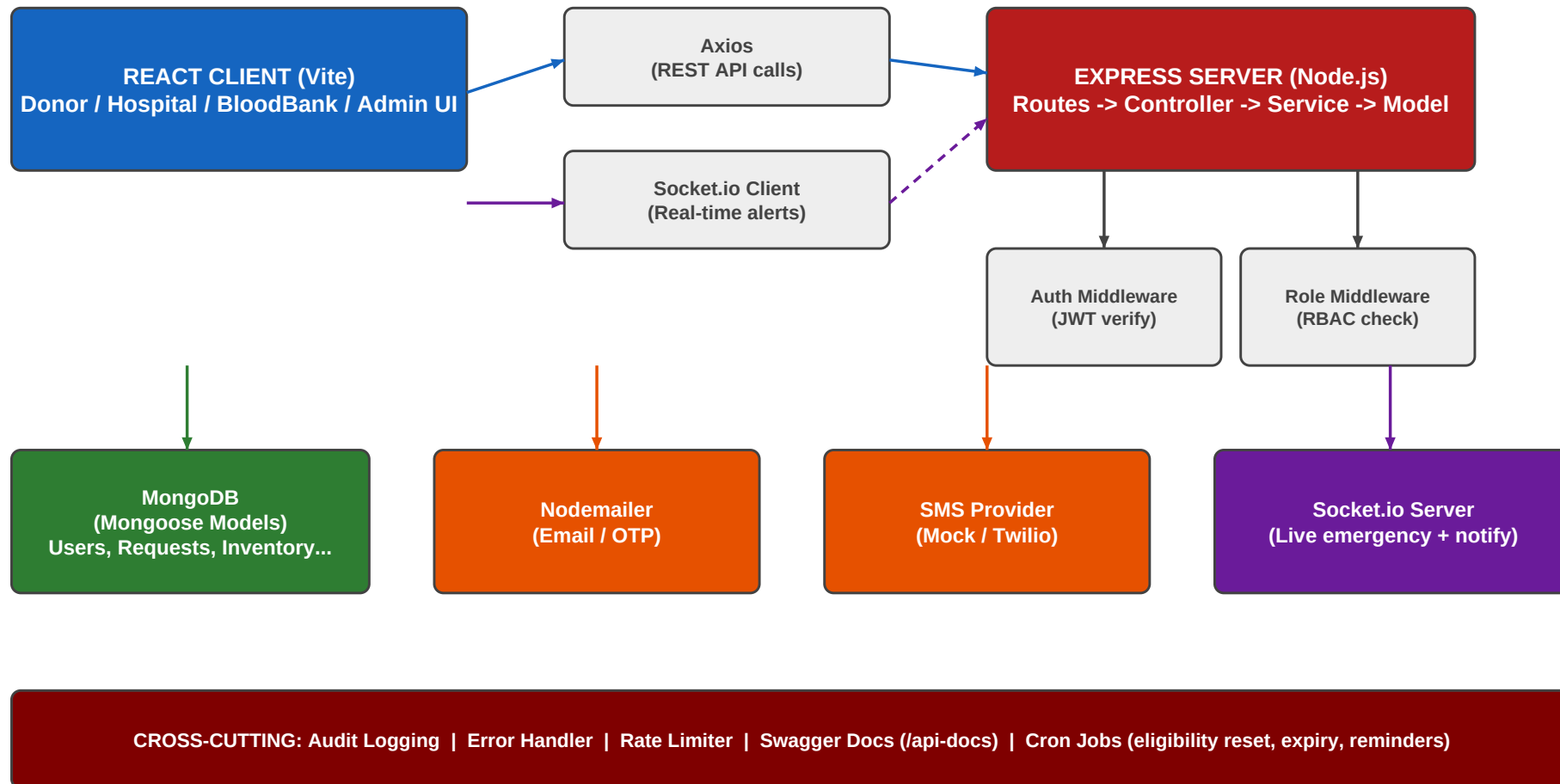


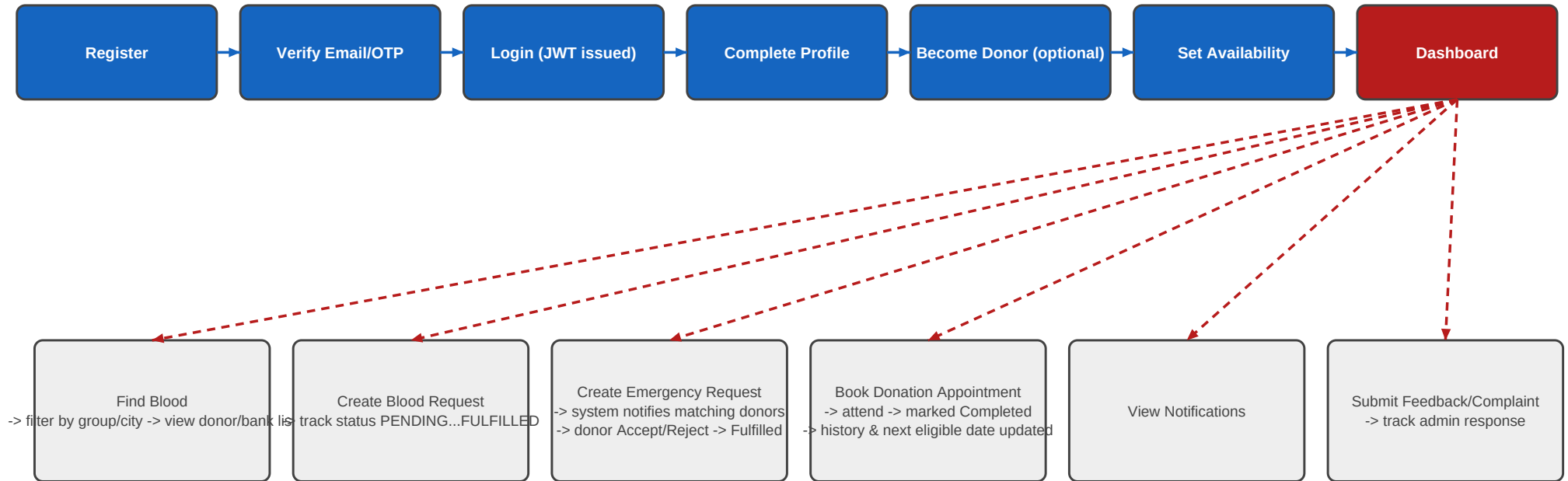
1. System Architecture (MERN Stack)

Kaise saare parts aapas me connect hote hain



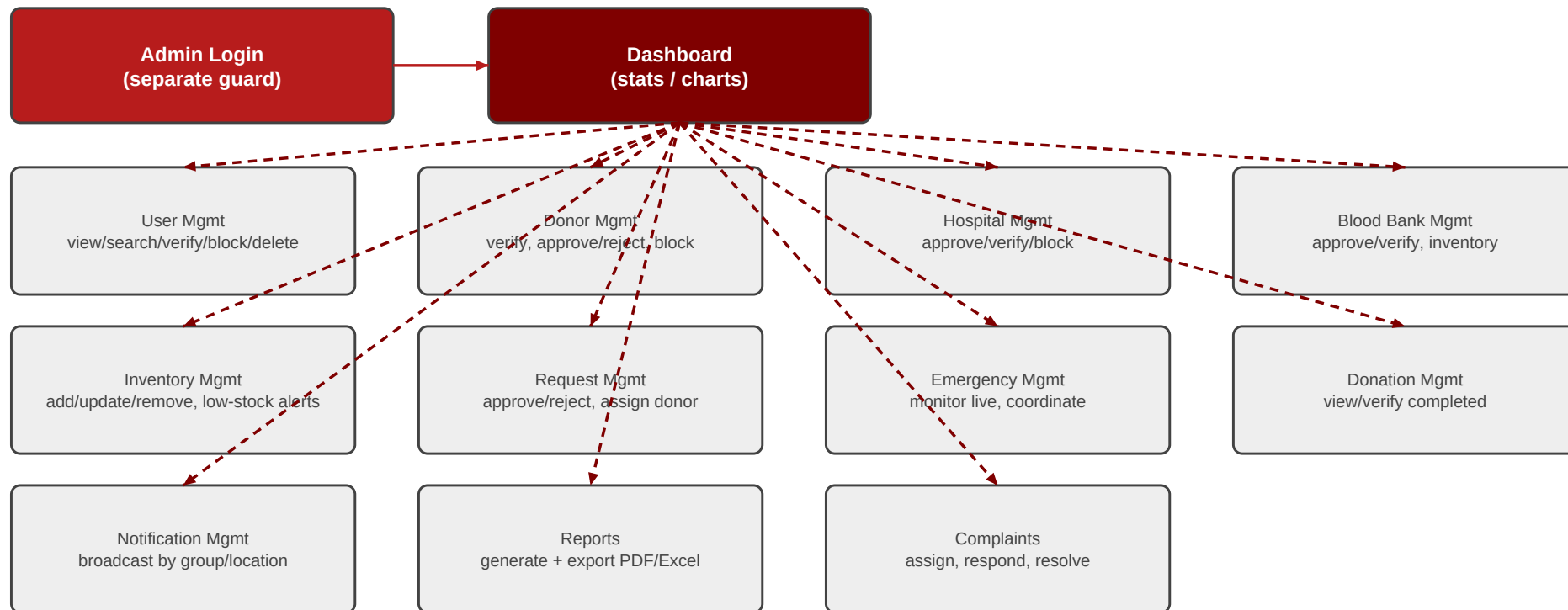
2. Donor / User Flow

Register se Dashboard tak ka safar



3. Admin Flow

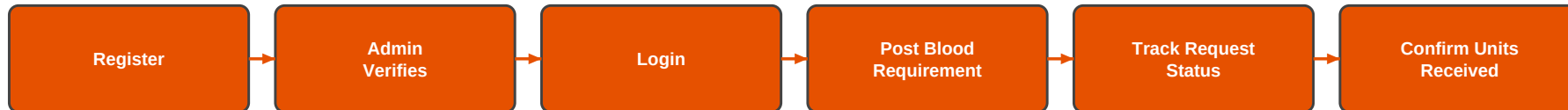
Admin login se saare management modules tak



4. Hospital & Blood Bank Flow

Registration se operations tak

HOSPITAL FLOW

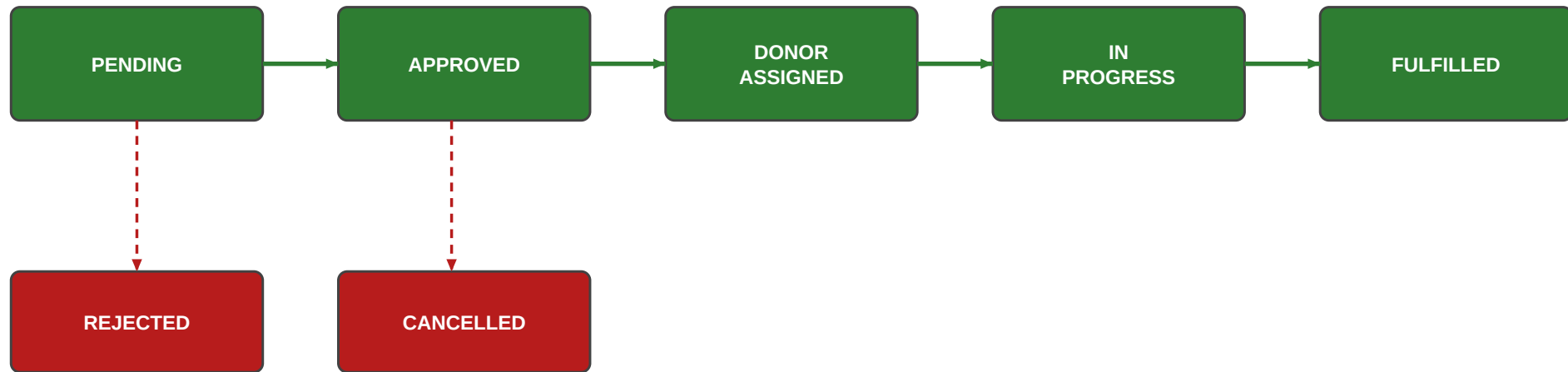


BLOOD BANK FLOW



5. Blood Request - Status Lifecycle

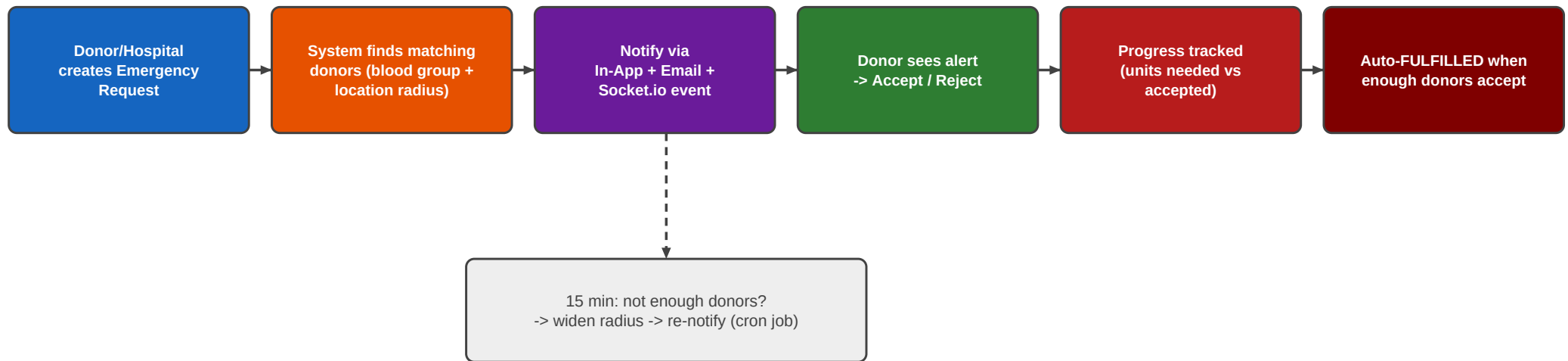
Ek request kis tarah ek state se dusri state me jaati hai



- Har state change 'statusHistory' array me record hota hai aur requester ko notification jaati hai.
- Invalid transition (jaise seedha PENDING -> FULFILLED) backend me block hona chahiye.
- REJECTED/CANCELLED kisi bhi PENDING/APPROVED state se ho sakta hai, FULFILLED ke baad nahi.

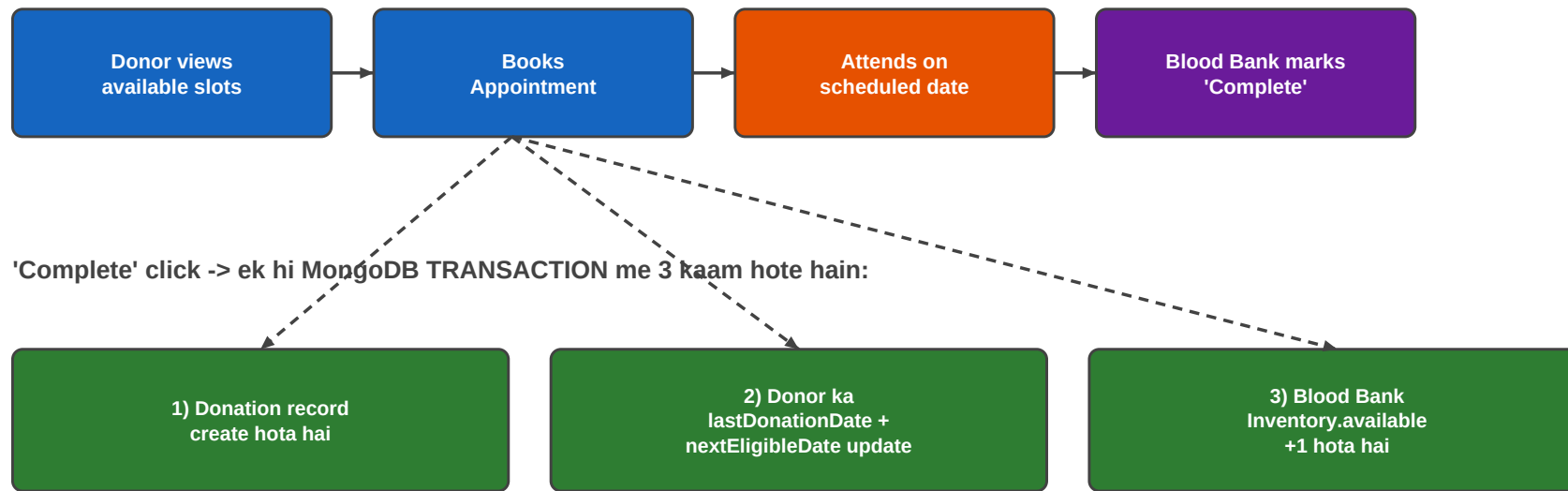
6. Emergency Request - Real-time Flow

Socket.io se instant donor matching



7. Appointment -> Donation Flow

Slot booking se donation record tak (transaction-safe)



- Agar donor nahi aata -> 'No-show' marked, koi donation record nahi banta.
- 'next eligible date' donor ke gender rule (90/120 din) ke hisaab se set hoti hai.